Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We use a fixed one-minute calendar-time grid from 09:30 to 15:59 for each trading day. This corresponds to 390 one-minute observations per normal trading day. Since a stock may not trade in every minute, we handle missing prices using previous-tick interpolation within each trading day only. This means that if AAPL has no observed price in a given minute, we carry forward the most recently observed price from the same trading day. We do not carry prices across trading days, and we do not backfill prices before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, we sample prices on a fixed grid relative to the market open. For example, with 10-minute sampling, prices are taken at 09:30, 09:40, 09:50, and so on. This differs from a bucket-based approach, where the last observed price within each Δ-minute interval is used. We prefer the fixed-grid approach because it makes the sampling rule explicit and because it will later allow us to align intraday returns across all 30 stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where (r_{d,j}^{( )}) is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

The volatility signature plot shows that the average realized variance is highest at the shortest sampling intervals and falls substantially when moving from 1-minute to 5-minute sampling. This is

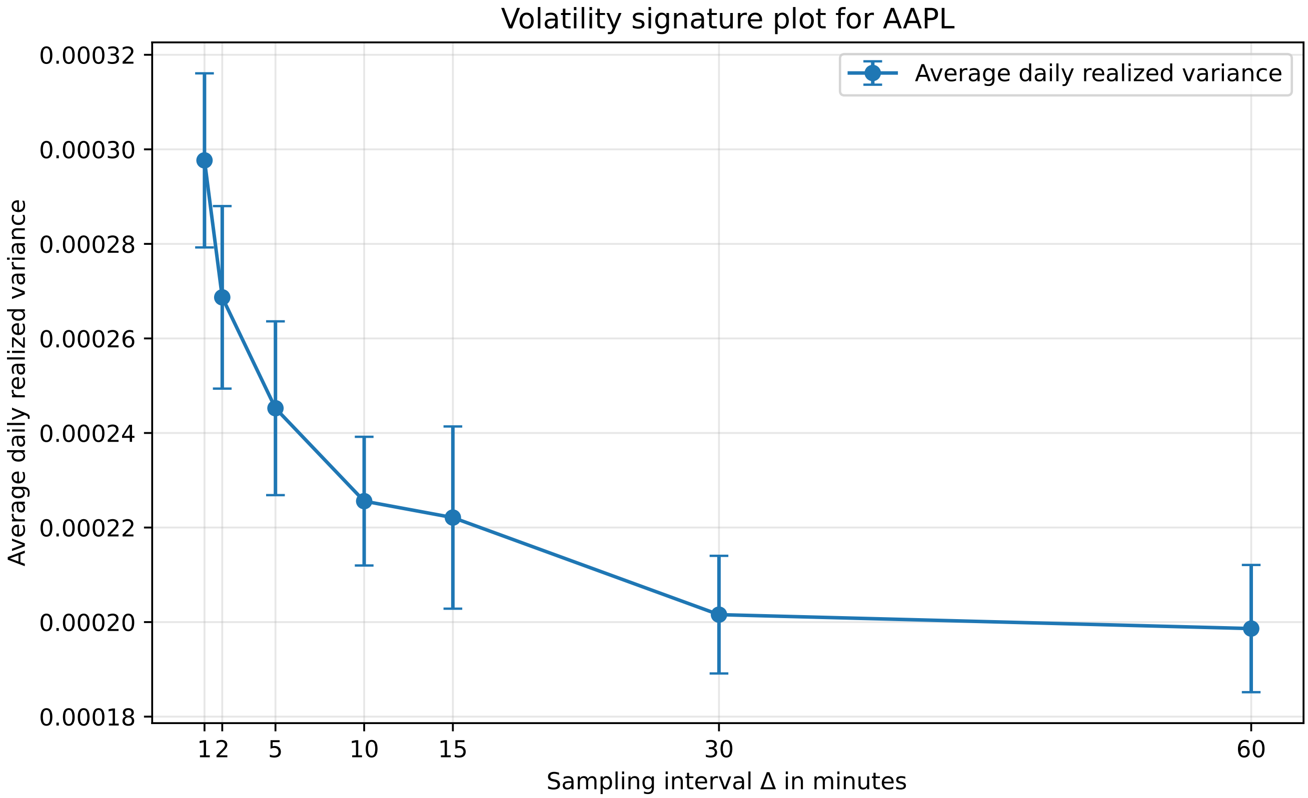

Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

consistent with the presence of market microstructure noise at very high frequencies. In the lecture slides, the observed log price is written as the latent efficient log price plus a noise component, $Y_{i\Delta_n} = X_{i\Delta_n} + U_{i\Delta_n}$, and the noise-driven fraction of return variance increases as Δ_n becomes small. Sampling less frequently therefore reduces the influence of bid-ask bounce, price discreteness and stale prices.

We choose the standard realized variance estimator based on 5-minute log returns for the remainder of the analysis:

$$RV_d^{(5)} = \sum_j \left(r_{d,j}^{(5)} \right)^2 .$$

This choice is also consistent with the empirical realized-volatility literature, where 5-minute returns are commonly used as a practical compromise between using high-frequency information and limiting market microstructure noise. Andersen and Bollerslev discuss the use of high-frequency intraday returns to obtain improved volatility measurements and refer to empirical analysis based on five-minute returns. At the same time, 5-minute sampling leaves approximately $390/5 - 1 = 77$ intraday returns per trading day. Since the subsequent realized covariance matrix is 30-dimensional, this helps avoid the rank problem emphasized in the lecture slides: when the return matrix has $T < N$, the sample covariance matrix is singular and cannot be inverted. Thus, 5-minute sampling is preferred to lower-frequency choices such as 15, 30 or 60 minutes, which would leave fewer intraday returns for estimating the 30×30 covariance matrix.

Exercise 2a: Daily realized covariance matrices

We estimate the daily realized covariance matrix for the 30 DJIA stocks using the standard realized covariance estimator

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j},$$

where $r_{d,j}$ denotes the 30×1 vector of synchronized intraday log returns at interval j on trading day d. This estimator is the multivariate analogue of the realized variance estimator used in Exercise 1.

Following Exercise 1, covariance is estimated using synchronized 5-minute returns. The same sampling frequency is used throughout the remainder of the analysis, ensuring consistency with the realized variance estimator selected based on the volatility signature plot.

To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day and missing observations are handled using previous-tick interpolation (forward filling). Returns are then computed on a common 5-minute grid, ensuring that the covariance matrix is based on synchronized return vectors.

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i, the diagonal element is

$$RC_{d,ii} = \sum_{j=1}^{M_d} r^2_{d,j,i}.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

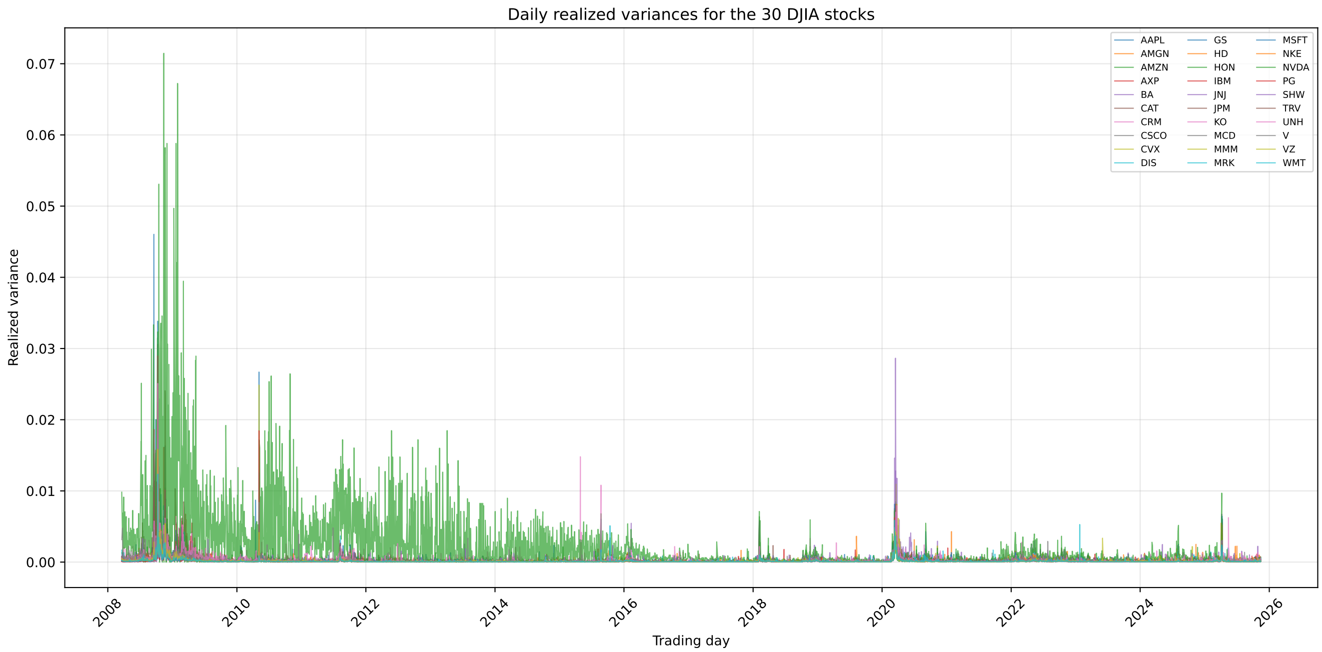

Figure 2: Daily realized variances for the 30 DJIA stocks.

The realized variance series exhibit clear volatility clustering, with periods of high volatility followed by further periods of high volatility and vice versa. Several stocks experience large volatility spikes during periods of market stress, indicating that volatility tends to increase simultaneously across assets.

The figure also shows substantial time variation in realized variance, suggesting that volatility is far from constant over time. While the overall patterns are similar across stocks, some stocks display persistently higher realized variance than others, reflecting differences in underlying risk characteristics.

References